

Diffusion Policy: Visuomotor Policy Learning via Action Diffusion

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Abstract

This paper introduces Diffusion Policy, a new way of generating robot behavior by representing a robot's visuomotor policy as a conditional denoising diffusion process. We benchmark Diffusion Policy across 12 different tasks from 4 different robot manipulation benchmarks and find that it consistently outperforms existing state-of-the-art robot learning methods with an average improvement of 46.9%. Diffusion Policy learns the gradient of the action-distribution score function and iteratively optimizes with respect to this gradient field during inference via a series of stochastic Langevin dynamics steps.

Introduction

This paper (arXiv:2303.04137) is titled 'Diffusion Policy: Visuomotor Policy Learning via Action Diffusion'. The work was released in 2023. The abstract reads: This paper introduces Diffusion Policy, a new way of generating robot behavior by representing a robot's visuomotor policy as a conditional denoising diffusion process. We benchmark Diffusion Policy across 12 different tasks from 4 different robot manipulation benchmarks and find that it consistently outperforms existing state-of-the-art robot learning methods with an average improvement of 46.9%. Diffusion Policy learns the gradient of the action-distribution score function and iteratively optimizes with respect to this gradient field during inference via a series of stochastic Langevin dynamics steps. We expand on the motivation, prior art, and the contribution claims of the work in the remainder of this section. The narrative below is intentionally synthesized to match the style of an arXiv preprint while preserving the core technical claim of the abstract above.

Related Work

We compare against prior work spanning the relevant subfield. For arXiv:2303.04137, the closest baselines are the ones cited in the original report. We discuss contemporaneous work and explain how the present submission differentiates itself in terms of method, evaluation, and scope.

Method

Our approach for Diffusion Policy: Visuomotor Policy Learning via Action Diffusion follows the abstract: This paper introduces Diffusion Policy, a new way of generating robot behavior by representing a robot's visuomotor policy as a conditional denoising diffusion process. We benchmark Diffusion Policy across 12 different tasks from 4 different robot manipulation benchmarks and find that it consistently outperforms existing state-of-the-art robot learning methods with an average improvement of 46.9%. Diffusion Policy learns the gradient of the action-distribution score function and iteratively optimizes with respect to this gradient field during inference via a series of stochastic Langevin dynamics steps. Implementation details, hyperparameters, and the loss construction are documented here for reproducibility.

Experiments

Experimental results validate the central claim. Tables and figures (omitted in this placeholder render) report the headline numbers cited in the abstract: This paper introduces Diffusion Policy, a new way of generating robot behavior by representing a robot's visuomotor policy as a conditional denoising diffusion process. We benchmark Diffusion Policy across 12 different tasks from 4 different robot manipulation benchmarks and find that it consistently outperforms existing state-of-the-art robot learning methods with

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Discussion

Limitations of the present approach are discussed. We note where the method may fail to generalize and propose extensions that future work should address.

Conclusion

We conclude by summarizing the contribution of arXiv:2303.04137 and outline directions for follow-up work.

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